

Literature status: connectivity and generic full spark for FUNTFs

Source arXiv:1110.3548; answer arXiv:1311.4748

13 August 2026

Status: literature already answered. This packet records a later, source-aware resolution; it does not claim a new proof.

Source question

In Section 3 of *Full Spark Frames*, Alexeev, Cahill, and Mixon [1] seek density of full-spark unit-norm tight frames (FUNTFs). On page 19 of the source PDF they write:

“It is unknown whether the variety of $M \times N$ unit norm tight frames is connected in general.”

They explain that connectedness, together with nonsingularity where available, would yield irreducibility and hence the desired density. At that point they can only show that every FUNTF is close to a full-spark tight frame whose column norms are close to one.

Decisive later answer

Cahill, Mixon, and Strawn [2] explicitly cite *Full Spark Frames* and state that they completely resolve the frame-homotopy problem. In their notation $\mathcal{F}_{N,d}^{\mathbb{R}}$ is the space of N -element FUNTFs in $\mathbb{F}^d$. On supporting-PDF page 2, Theorems 1.1 and 1.2 prove

$\mathcal{F}_{N,d}^{\mathbb{C}}$ is path-connected for every $N \geq d$,

and

$\mathcal{F}_{N,d}^{\mathbb{R}}$ is path-connected for $d \geq 2$, $N \geq d + 2$.

The low-redundancy qualification is real: the same paper records that $\mathcal{F}_{d+1,d}^{\mathbb{R}}$ has 2^d connected components. Thus the source’s unrestricted real connectedness question is false, while the whole genuinely redundant range $N \geq d + 2$ is affirmative. The complex question is affirmative without exceptions.

More directly, supporting-PDF page 3, Theorem 1.6, proves that a generic frame in $\mathcal{F}_{N,d}^{\mathbb{C}}$ and in $\mathcal{F}_{N,d}^{\mathbb{R}}$ is full spark for every $N \geq d$. This settles the density objective that motivated the source question, including the real exceptional spaces where connectedness itself fails.

Provenance, scope, and search

This is an explicit literature answer rather than an agent-derived implication: the supporting paper cites Alexeev–Cahill–Mixon, uses its definition of full spark, and announces both a complete

homotopy resolution and generic full spark. The packet does not address unrelated deterministic RIP or complexity questions elsewhere in the source paper.

The run registry and local sources were searched for arXiv:1110.3548, “unit norm tight frames connected,” “FUNTF connectivity,” and “full spark.” A bounded web search through 13 August 2026 located the decisive arXiv:1311.4748 paper and later symplectic/topological confirmations.

References

- [1] B. Alexeev, J. Cahill, and D. G. Mixon, *Full Spark Frames*, J. Fourier Anal. Appl. 18 (2012), 1167–1194; arXiv:1110.3548.
- [2] J. Cahill, D. G. Mixon, and N. Strawn, *Connectivity and Irreducibility of Algebraic Varieties of Finite Unit Norm Tight Frames*, SIAM J. Appl. Algebra Geom. 1 (2017), 38–72; arXiv:1311.4748.